Final Exam Review — Solutions

Calculus II · STM 1002 · Spring 2026

Solutions to the practice problems in `FinalReview_NotesAndProblems`. Methods matter more than the final number — compare your reasoning, not just your answer.

1. Accumulation & the FTC

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1. By FTC Part 1, $\dfrac{d}{dx}\displaystyle\int_2^x \sqrt{1+t^3}\,dt = \sqrt{1+x^3}$.

2. The variable is the lower limit, so flip: $\displaystyle\int_x^5 \sin(t^2)\,dt = -\int_5^x \sin(t^2)\,dt$, and the derivative is $-\sin(x^2)$.

3. (a) Additivity: $\displaystyle\int_0^5 f = \int_0^3 f + \int_3^5 f = 7 + 2 = 9$. (b) Reversal: $\displaystyle\int_5^0 f = -\int_0^5 f = -9$.

4. $\Delta t = 2$. **Left** ($t = 0, 2, 4$): $(5 + 8 + 6)(2) = 38$. **Right** ($t = 2, 4, 6$): $(8 + 6 + 3)(2) = 34$. The integral is the total volume (in liters) that flowed during the 6 minutes; units: $(\text{L/min})\times\text{min} = \text{L}$.

5. $F' = f$. So F is **decreasing on** $(0, 1)$ (where $f < 0$) and **increasing on** $(1, 4)$ (where $f > 0$). F' changes from negative to positive at $x = 1$, so F has a **local minimum at** $x = 1$.

6. Net displacement $= \displaystyle\int_0^5 (12 - 3t)\,dt = \left[12t - \tfrac{3}{2}t^2\right]_0^5 = 60 - 37.5 = \mathbf{22.5}$ m. The velocity is positive until $t = 4$ and negative on $(4, 5]$, so the car backs up at the end; that reversed motion cancels part of the forward progress, making net displacement (22.5 m) smaller than the total distance driven (25.5 m).

2. Antiderivatives & Techniques

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7. $\displaystyle\int (6x^2 - 4x + 1)\,dx = 2x^3 - 2x^2 + x + C$. $\displaystyle\int \cos(3x)\,dx = \tfrac{1}{3}\sin(3x) + C$.

8. (a) $u = x^2$: $\tfrac{1}{2}e^{x^2} + C$. (b) $u = \ln x$: $\tfrac{1}{2}(\ln x)^2 + C$. (c) $u = x^2 + 1$: $\tfrac{1}{2}\ln(x^2 + 1)\big|_0^1 = \tfrac{1}{2}\ln 2 \approx 0.347$.

9. (a) $u = x$, $dv = e^x dx$: $\displaystyle\int xe^x\,dx = xe^x - e^x + C = e^x(x - 1) + C$. (b) $u = x$, $dv = \sin x\,dx$: $-x\cos x + \sin x + C$. (c) $u = \ln x$, $dv = dx$: $x\ln x - x + C$.

10. $v = t^2 - 4$, zero at $t = 2$ (negative on $[0, 2)$, positive on $(2, 3]$). (a) Net $= \displaystyle\int_0^3 (t^2 - 4)\,dt = [\tfrac{t^3}{3} - 4t]_0^3 = 9 - 12 = \mathbf{-3}$ m. (b) Total $= \displaystyle\int_0^2 (4 - t^2)\,dt + \int_2^3 (t^2 - 4)\,dt = \tfrac{16}{3} + \tfrac{7}{3} = \tfrac{23}{3} \approx \mathbf{7.67}$ m.

11. $\int x^2\sqrt{x^3+1}\,dx$: **substitution** $u = x^3+1$ ($du = 3x^2\,dx$ present). $\int x^2 \ln x\,dx$: **parts** (polynomial $\times$ ln; differentiate $\ln x$). $\int \frac{x}{1+x^2}\,dx$: **substitution** $u = 1+x^2$ (f'/f). $\int \sin x\cos x\,dx$: **substitution** $u = \sin x$ (its derivative $\cos x$ is present).

3. Applications: Area, Surplus, Series, Present Value

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12. On $[0,1]$, $\sqrt{x} \geq x$. Area $= \displaystyle\int_0^1 (\sqrt{x}-x)\,dx = \left[\tfrac{2}{3}x^{3/2} - \tfrac{x^2}{2}\right]_0^1 = \tfrac{2}{3} - \tfrac{1}{2} = \boldsymbol{\tfrac{1}{6}}$.

13. Intersect: $4-x^2 = x+2 \Rightarrow x^2+x-2 = 0 \Rightarrow x = -2, 1$; the parabola is on top. Area $= \displaystyle\int_{-2}^{1} (2-x-x^2)\,dx = \left[2x - \tfrac{x^2}{2} - \tfrac{x^3}{3}\right]_{-2}^{1} = \tfrac{7}{6} - (-\tfrac{10}{3}) = \boldsymbol{\tfrac{9}{2}}$.

14. CS $= \displaystyle\int_0^5 \big[(20-2q)-10\big]\,dq = \int_0^5 (10-2q)\,dq = [10q - q^2]_0^5 = 50 - 25 = \mathbf{25}$.

15. $\displaystyle\sum_{k=0}^{\infty} 5\left(\tfrac{1}{3}\right)^k = \frac{5}{1-\frac{1}{3}} = \frac{5}{2/3} = \boldsymbol{\tfrac{15}{2}}$. And $0.\overline{27} = \dfrac{27/100}{1-1/100} = \dfrac{27}{99} = \boldsymbol{\tfrac{3}{11}}$.

16. (a) PV $= \dfrac{8000(1-e^{-0.5})}{0.05} = \dfrac{8000(0.3935)}{0.05} \approx \$\mathbf{62{,}960}$. (b) Perpetuity: $\dfrac{C}{r} = \dfrac{8000}{0.05} = \$\mathbf{160{,}000}$. (c) Each future dollar is discounted by $e^{-rt} \to 0$, so distant payments contribute almost nothing; the discounted total converges even though the payments never end.

4. The Normal Distribution

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17. $\mu = 50$, $\sigma = 10$. (a) $z = \frac{65-50}{10} = 1.5$: $P(X < 65) = \Phi(1.5) = \mathbf{0.9332}$. (b) $z = \frac{40-50}{10} = -1$: $P(X > 40) = P(Z > -1) = \Phi(1) = \mathbf{0.8413}$. (c) $z = \pm 1$: $\Phi(1) - \Phi(-1) = 0.8413 - 0.1587 = \mathbf{0.6826}$.

18. Above $\mu + \sigma$: $(1-0.68)/2 = \mathbf{16}\%$. Above $\mu - 2\sigma$: everything except the lower 2.5% tail $= \mathbf{97.5}\%$.

19. $[19, 21] = 20 \pm 2(0.5) = \mu \pm 2\sigma$, which holds $\approx 95\%$ of the mass, so $\approx$ **5**% is scrapped. One vulnerable assumption: that lengths are *exactly* normal with the mean fixed at 20 and σ constant — tool wear, drift, or a skewed defect mode would break the empirical-rule estimate.

5. Taylor Series

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20. (a) $\sin x: \; T_3 = x - \dfrac{x^3}{6}$. (b) $\dfrac{1}{1-x}: \; T_3 = 1 + x + x^2 + x^3$.

21. $T_2 = 1 + 0.2 + \dfrac{(0.2)^2}{2} = \mathbf{1.22}$. Lagrange: $|R_2| \leq \dfrac{\max e^c\,(0.2)^3}{3!} \leq \dfrac{1.25(0.008)}{6} \approx \mathbf{1.7 \times 10^{-3}}$

(true value 1.2214, error $\approx 1.4 \times 10^{-3}$, within the bound). Lagrange is the right tool because the series for e^x has *all positive* terms — it is not alternating, so the next-term bound does not apply.

22. $\int_0^1 \sin(x^2)\,dx \approx \int_0^1 \left(x^2 - \frac{x^6}{6}\right) dx = \frac{1}{3} - \frac{1}{42} = \frac{14-1}{42} = \mathbf{\frac{13}{42}} \approx \mathbf{0.3095}$.

23. Terms $\frac{(0.1)^k}{k}$: $0.1,\ 0.005,\ 3.33 \times 10^{-4},\ 2.5 \times 10^{-5}$. The first term below 10^{-4} is the degree-4 term, so keep through degree 3 — **3 terms**: $T_3(0.1) = 0.1 - 0.005 + 0.000333 \approx \mathbf{0.09533}$ (true $\ln 1.1 \approx 0.09531$).

24. $\sin\theta - \theta$ is bounded by the next term: $\left|\sin(0.3) - 0.3\right| \le \dfrac{(0.3)^3}{6} = \dfrac{0.027}{6} = \mathbf{4.5 \times 10^{-3}}$. It is accurate for small θ because the omitted $\theta^3/6$ term is tiny when θ is small; for large θ that cubic (and higher) terms grow and the linear approximation falls apart.

25. (a) At $x = \frac{1}{2}$: $\sum(\frac{1}{2})^k$ **converges** to $\dfrac{1}{1-\frac{1}{2}} = \mathbf{2}$. At $x = 2$: the terms 2^k grow without bound, so the series **diverges**. (b) The radius is the distance from the center ($x = 0$) to the nearest point where $\ln(1+x)$ misbehaves: the **singularity at** $x = -1$, where $1 + x = 0$ and ln blows up. That distance is 1, so the series converges only for $|x| < 1$.

6. Differential Equations & Dynamics

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26. $y = 4 + Ce^{-3t} \Rightarrow y' = -3Ce^{-3t}$. And $-3(y-4) = -3(Ce^{-3t}) = -3Ce^{-3t}$. The two sides agree for every C. ✓

27. (a) Separate: $\dfrac{dy}{y} = 3x\,dx \Rightarrow \ln|y| = \frac{3}{2}x^2 + C \Rightarrow y = Ae^{3x^2/2}$. With $y(0) = 2$, $A = 2$: $y = \mathbf{2e^{3x^2/2}}$. (b) $\dfrac{dy}{y^2} = dx \Rightarrow -\frac{1}{y} = x + C$. With $y(0) = 1$, $C = -1$: $-\frac{1}{y} = x - 1 \Rightarrow y = \dfrac{1}{1-x}$. It **blows up as** $x \to 1$ (finite-time blow-up).

28. $f(y) = y(4-y)$, fixed points $y = 0$ and $y = 4$. $f'(y) = 4 - 2y$: $f'(0) = 4 > 0$ (**unstable**), $f'(4) = -4 < 0$ (**stable**). Starting at $y(0) = 1$ (between 0 and 4, where $f > 0$), the solution increases and approaches the stable level $y = 4$.

29. (a) With $h = 0$: $P' = 0.5P(1 - P/1000)$, fixed points $P = 0$ (unstable) and $P = 1000$ (stable carrying capacity). (b) As h increases, the upper (stable) equilibrium drops below 1000 and the lower (unstable) one rises; they collide at the peak of the growth curve. The maximum sustainable harvest is $h_{\max} = \dfrac{rK}{4} = \dfrac{0.5 \cdot 1000}{4} = 125$. For $h > 125$ there is *no* positive equilibrium — $P' < 0$ everywhere — and the population collapses to zero (over-harvesting).